SIMATS ENGINEERING

ASSESSMENT TOOL 1 - ANALYTICAL PROBLEM SOLVING

Course Name	Computer Networks
Course Code	CSA0709
Course Outcome (CO2)	Configure IP addressing schemes, subnetting, and basic routing techniques for efficient network design and topology (BL3)
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Assessment Tool Weightage	40%

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Question 1 (8 Marks)

Question: A university network is assigned the IP address block 192.168.20.0/24. It is divided into subnets for three labs using fixed subnet masks: Lab A = /26, Lab B = /27, Lab C = /28. Determine the network address of each lab, the valid host range, the broadcast address, and the number of usable hosts in each subnet.

1. Problem Understanding

- Given block: 192.168.20.0/24 (256 addresses).
- Required: network address, host range, broadcast address and usable hosts for Lab A (/26), Lab B (/27) and Lab C (/28).

2. Method Selection

- Host bits h = 32 - prefix; block size = 2^h; usable hosts = 2^h - 2.
- Network address = first address of block; broadcast = last address; hosts lie between them.
- Subnets are allocated one after another from the start of the block.

3. Solution Process

Step 1 - Masks and block sizes

Lab	Prefix	Subnet Mask	Host bits	Block size	Usable hosts
A	/26	255.255.255.192	6	64	2^6 - 2 = 62
B	/27	255.255.255.224	5	32	2^5 - 2 = 30
C	/28	255.255.255.240	4	16	2^4 - 2 = 14

Step 2 - Sequential allocation: Lab A: .0 to .63; Lab B starts at the next free address .64 and ends at .95; Lab C starts at .96 and ends at .111.

4. Final Answer

Lab	Network Address	Valid Host Range	Broadcast	Usable Hosts
A (/26)	192.168.20.0	192.168.20.1 - 192.168.20.62	192.168.20.63	62
B (/27)	192.168.20.64	192.168.20.65 - 192.168.20.94	192.168.20.95	30
C (/28)	192.168.20.96	192.168.20.97 - 192.168.20.110	192.168.20.111	14

5. Interpretation of Results

Lab A supports up to 62 devices, Lab B 30 and Lab C 14. The network and broadcast addresses of each subnet cannot be given to hosts. The three subnets use 112 addresses (192.168.20.0 - 192.168.20.111), so 192.168.20.112 - 192.168.20.255 remains free for future subnets.

Question 2 (8 Marks)

Question: System 1 has IP address 192.168.1.130 and System 2 has IP address 192.168.1.190, both with subnet mask 255.255.255.192. Determine the subnet range, network address and broadcast address for each system, the valid host range, whether each IP falls within its subnet, and the number of usable hosts.

1. Problem Understanding

- Mask 255.255.255.192 = /26 (2 bits borrowed in the last octet).
- Find the subnet of each IP and verify that each address lies within the subnet range.

2. Method Selection

- Block size = 256 - 192 = 64, so subnets start at 0, 64, 128, 192.
- Network address = IP AND subnet mask.
- Usable hosts = 2^6 - 2 = 62.

3. Solution Process

System 1 - 192.168.1.130: 130 = 10000010; mask 192 = 11000000; AND = 10000000 = 128. Network = 192.168.1.128; broadcast = 128 + 64 - 1 = 191; hosts = .129 to .190.

System 2 - 192.168.1.190: 190 = 10111110; mask 192 = 11000000; AND = 10000000 = 128. Network = 192.168.1.128; broadcast = 191; hosts = .129 to .190.

4. Final Answer

Parameter	System 1	System 2
IP address	192.168.1.130	192.168.1.190
Subnet mask	255.255.255.192 (/26)	255.255.255.192 (/26)
Subnet range	192.168.1.128 - 192.168.1.191	192.168.1.128 - 192.168.1.191
Network address	192.168.1.128	192.168.1.128
Broadcast address	192.168.1.191	192.168.1.191
Valid host range	192.168.1.129 - 192.168.1.190	192.168.1.129 - 192.168.1.190
Within its subnet?	Yes (129 <= 130 <= 190)	Yes (129 <= 190 <= 190)
Usable hosts	62	62

5. Interpretation of Results

Both systems belong to the same subnet (192.168.1.128/26), so they can communicate directly without a router. 192.168.1.190 is the last usable host address; the next address, 192.168.1.191, is the broadcast address. The subnet supports 62 hosts.

Question 3 (8 Marks)

Question: An organization is allocated 172.16.0.0/16. Department masks: HR = /25, Sales = /26, IT = /27, Admin = /28. Assign subnet addresses, and determine the network address, valid host range, broadcast address and usable hosts for each, and the remaining unused address range.

1. Problem Understanding

- Parent block: 172.16.0.0/16 (65,536 addresses).
- Four departments with fixed masks need non-overlapping subnets; the unused range must be identified.

2. Method Selection

- Block size = $2^{(32 - \text{prefix})}$; usable hosts = block size - 2.
- Allocate largest to smallest, one after another, so each subnet is aligned and nothing overlaps.

3. Solution Process

Step 1 - Block sizes

Department	Prefix	Mask	Block size	Usable hosts
HR	/25	255.255.255.128	128	126
Sales	/26	255.255.255.192	64	62
IT	/27	255.255.255.224	32	30
Admin	/28	255.255.255.240	16	14

Step 2 - Allocation (all inside 172.16.0.x): HR 0-127; Sales 128-191; IT 192-223; Admin 224-239.

4. Final Answer

Department	Network Address	Valid Host Range	Broadcast	Usable Hosts
HR	172.16.0.0/25	172.16.0.1 - 172.16.0.126	172.16.0.127	126
Sales	172.16.0.128/26	172.16.0.129 - 172.16.0.190	172.16.0.191	62
IT	172.16.0.192/27	172.16.0.193 - 172.16.0.222	172.16.0.223	30
Admin	172.16.0.224/28	172.16.0.225 - 172.16.0.238	172.16.0.239	14

Unused range: 172.16.0.240 - 172.16.255.255. Addresses used = 128 + 64 + 32 + 16 = 240; unused = 65,536 - 240 = **65,296**.

5. Interpretation of Results

All four departments fit in the first 240 addresses with no overlap. Total usable hosts = 126 + 62 + 30 + 14 = 232. Almost the whole /16 block remains available for future departments or subnets.

Question 4 (8 Marks)

Question: A company uses 172.16.10.0/24 with a /27 subnet mask. Calculate the total number of subnets and usable hosts per subnet. Find the subnet of 172.16.10.78, its network and broadcast addresses, and determine whether 172.16.10.95 falls in the same subnet.

1. Problem Understanding

- Original network 172.16.10.0/24; new prefix /27, so 3 bits are borrowed from the host part.
- Need subnet count, hosts per subnet, and subnet membership of two addresses.

2. Method Selection

- Subnets = 2^n, n = 27 - 24 = 3. Usable hosts = 2^h - 2, h = 32 - 27 = 5.
- Block size = 256 - 224 = 32; network address = IP AND mask.

3. Solution Process

Step 1: Subnets = 2^3 = **8**. Usable hosts per subnet = 2^5 - 2 = **30**. Mask = 255.255.255.224.
Step 2 - Subnet table (block size 32):

Subnet	Network	Host range	Broadcast
1	172.16.10.0	.1 - .30	.31
2	172.16.10.32	.33 - .62	.63
3	172.16.10.64	.65 - .94	.95
4	172.16.10.96	.97 - .126	.127
5	172.16.10.128	.129 - .158	.159
6	172.16.10.160	.161 - .190	.191
7	172.16.10.192	.193 - .222	.223
8	172.16.10.224	.225 - .254	.255

Step 3 - 172.16.10.78: 78 = 01001110; mask 224 = 11100000; AND = 01000000 = 64. Network = 172.16.10.64; broadcast = 64 + 32 - 1 = 95; hosts = .65 to .94.

Step 4 - 172.16.10.95: 95 = 01011111; AND 11100000 = 01000000 = 64, so it is in the same subnet - but it is that subnet's broadcast address.

4. Final Answer

- Total subnets = **8**; usable hosts per subnet = **30**.
- 172.16.10.78 belongs to subnet **172.16.10.64/27**.
- Network address = **172.16.10.64**; broadcast address = **172.16.10.95**.
- 172.16.10.95 is within the same subnet range, but being the broadcast address it **cannot be assigned to a host**.

5. Interpretation of Results

The /27 mask splits the /24 network into eight equal subnets of 32 addresses, giving 30 usable hosts each. Host 172.16.10.78 is a valid host in the third subnet (172.16.10.64/27). Address 172.16.10.95 belongs to the same subnet range, but as the last address of the block it is used to send to all devices in that subnet, not to a single device.

Question 5 (8 Marks)

Question: A network uses the IPv6 address 2001:0db8:0000:0000:0000:ff00:0042:8329. Compress the address into its shortest form, then expand it back to full notation. Using a /64 prefix, determine the network prefix and the total number of possible host addresses.

1. Problem Understanding

- An IPv6 address has 128 bits, written as 8 groups of 4 hex digits separated by colons.
- Tasks: compression, expansion, /64 network prefix and number of host addresses.

2. Method Selection

- Rule 1: leading zeros in each group may be dropped.
- Rule 2: one (the longest) run of consecutive all-zero groups may be replaced by '::'.
- Network prefix = first 64 bits (first 4 groups); host bits = 128 - 64 = 64; total addresses = 2^{64}.

3. Solution Process

Step 1 - Compression: Full: 2001:0db8:0000:0000:0000:ff00:0042:8329. Drop leading zeros: 2001:db8:0:0:0:ff00:42:8329. Replace the zero groups (groups 3, 4, 5) with "::" giving **2001:db8::ff00:42:8329**.

Step 2 - Expansion: 2001:db8::ff00:42:8329 has 5 visible groups, so "::" stands for 8 - 5 = 3 zero groups. Insert 0:0:0 and pad each group to 4 digits: **2001:0db8:0000:0000:0000:ff00:0042:8329** (same as the original).

Step 3 - Network prefix (/64): first 64 bits = first four groups = 2001:0db8:0000:0000, i.e. **2001:db8::/64**. Interface ID (host part) = 0000:ff00:0042:8329.

Step 4 - Host addresses: host bits = 128 - 64 = 64, so total = 2^{64} = 18,446,744,073,709,551,616.

4. Final Answer

Item	Result
Compressed address	2001:db8::ff00:42:8329
Expanded address	2001:0db8:0000:0000:0000:ff00:0042:8329
Network prefix (/64)	2001:db8::/64
Host bits	64
Total host addresses	2^{64} = 18,446,744,073,709,551,616

5. Interpretation of Results

Compression shortens the address without changing its value, and expansion restores the exact original. A /64 prefix is the standard IPv6 subnet size: 64 bits for the network and 64 bits for the host. IPv6 has no broadcast address, so nothing is subtracted, giving about 18.4 quintillion addresses in one subnet.